

Cybersecurity Home Lab: Active Directory, Firewall Segmentation, Attack Simulation, & SIEM Logging

All the lab work is done in VMware Workstation Pro.

Overview

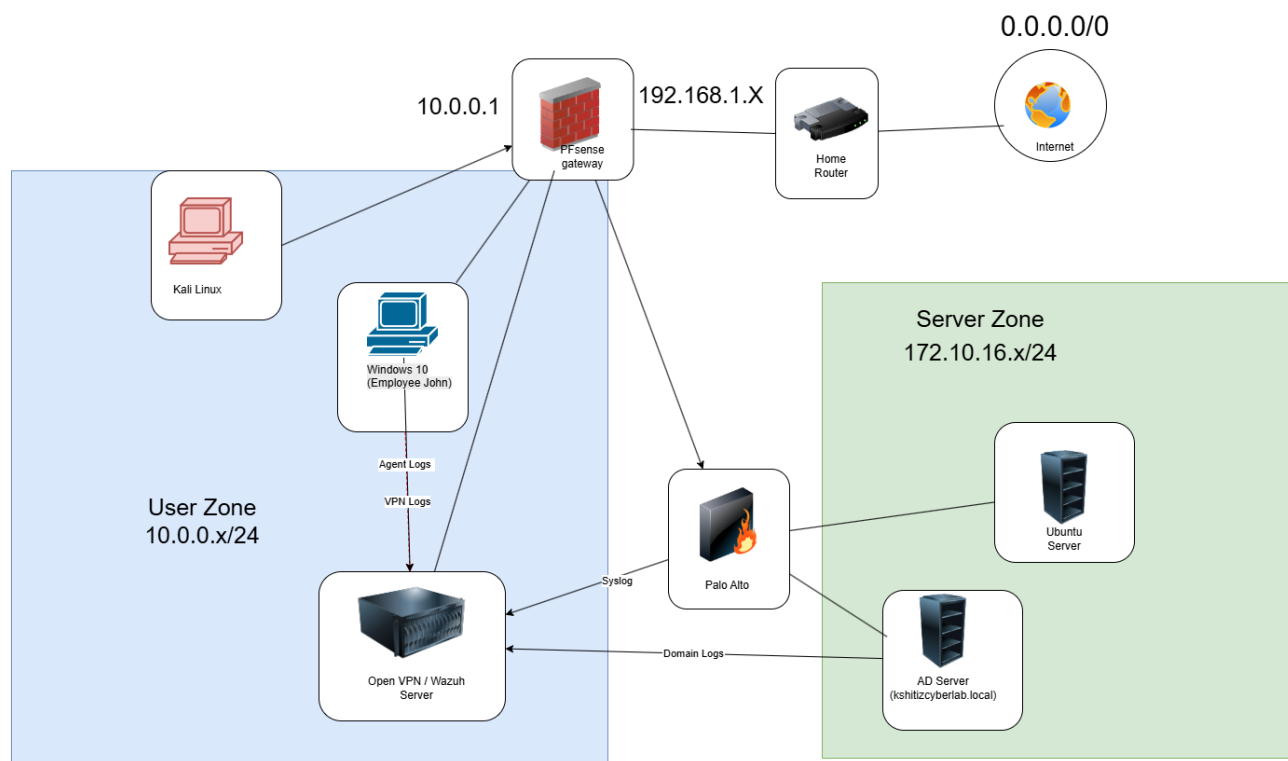
This project is a practical Security Operations Center (SOC) home lab designed to simulate real-world enterprise security monitoring. The Kali Linux machine is placed inside the user network (10.0.0.0/24). This represents a real-world scenario where an attacker has already gained access to the internal network, typically through phishing, malware, or stolen credentials.

The lab demonstrates how logs are generated, collected, and analyzed to detect security events such as:

- User account creation
- Privilege escalation
- SSH brute-force attempts
- Network scanning and enumeration

The environment combines endpoint monitoring, Active Directory logging, firewall inspection, and attack simulation.

Architecture



The lab is divided into multiple networks to simulate a real enterprise environment.

- **User Network (10.0.0.0/24):**
Contains endpoint systems and attacker machine
- **Server Network (172.16.10.0/24):**
Contains critical infrastructure like Active Directory and internal servers

Core Network Flow

Internet → Home Router → pfSense → Palo Alto Firewall → Server Network

All traffic to the server network must pass through the Palo Alto firewall.

🔧 Infrastructure & IP Configuration

Core Systems

Component	Role	IP Address
pfSense	Gateway	10.0.0.1
Palo Alto Firewall	Internal segmentation firewall	10.0.0.2 / 172.16.10.1
Wazuh Server (Ubuntu)	SIEM	10.0.0.60
Active Directory	Domain Controller	172.16.10.10

Windows 10 (John VM)	Endpoint	10.0.0.50
Kali Linux	Attacker	10.0.0.40
Ubuntu Server	Target (SSH/Web/FTP)	172.16.10.20

Interface Details

pfSense

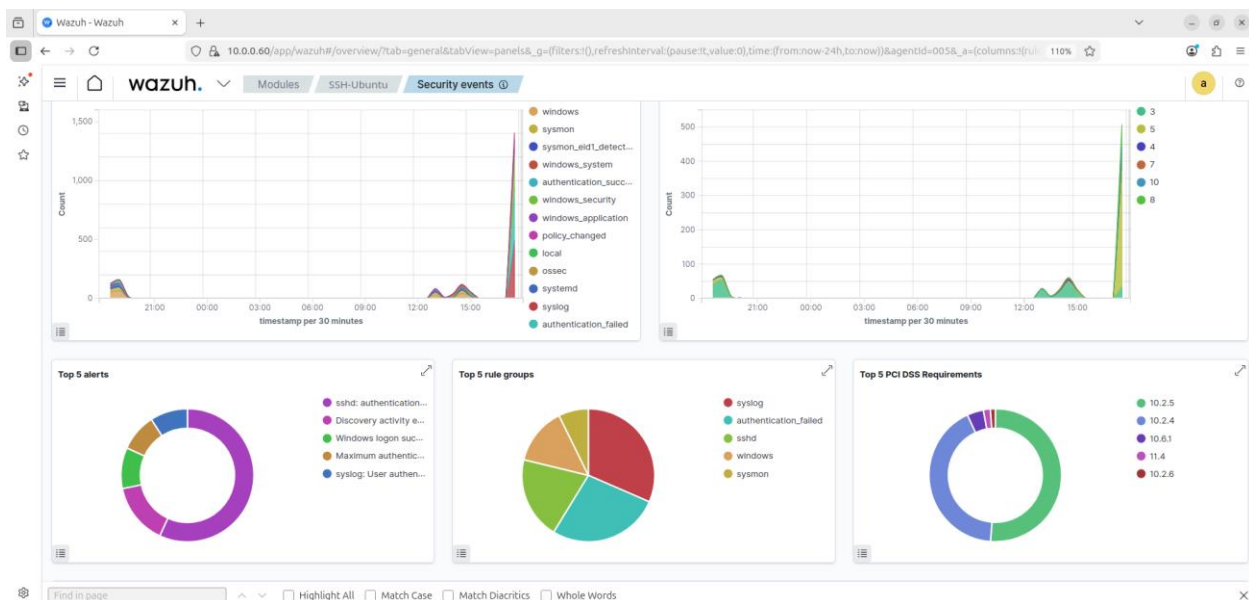
- WAN: 192.168.1.x (connected to home router)
- LAN: 10.0.0.1

Palo Alto Firewall

- eth1/1 (Untrust): 10.0.0.2
- eth1/2 (Trust): 172.16.10.1

🔒 Wazuh SIEM Configuration

The Wazuh SIEM was configured to collect logs from multiple sources.



Reliable Log Ingestion

The default UDP communication was replaced with TCP on port 1514.

This ensures that log messages are reliably delivered without packet loss.

```
GNU nano 7.2 /var/ossec/etc/ossec.conf *
<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>no</logall>
    <logall_json>no</logall_json>
    <email_notification>no</email_notification>
  </global>

  <alerts>
    <log_alert_level>3</log_alert_level>
    <email_alert_level>12</email_alert_level>
  </alerts>

  <remote>
    <connection>syslog</connection>
    <port>514</port>
    <protocol>udp</protocol>
    <allowed-ips>10.0.0.20</allowed-ips>
  </remote>

  <remote>
    <connection>secure</connection>
    <port>1514</port>
    <protocol>tcp</protocol>
    <queue_size>131072</queue_size>
  </remote>

  <localfile>
    <log_format>syslog</log_format>
    <location>/var/log/syslog</location>
  </localfile>
</ossec_config>
```

Syslog Integration

The Palo Alto firewall was configured to forward syslog to Wazuh

This allows firewall-level visibility inside the SIEM.

⚠ pfSense is not configured to forward syslog in this setup.

❑ Active Directory & Auditig

The domain kshitizcyberlab.local was created using Windows Server 2022.

🔥 Firewall & Network Security

The Palo Alto firewall is placed between the user network and the server network.

🌐 Target Server (Ubuntu) – Exposed Services

The Ubuntu server (172.16.10.20) was configured as an attack target.

Running services:

- SSH (22) – remote access
- FTP (21) – file transfer
- HTTP (80) – web server (Apache)
- MySQL (localhost only)

This creates a realistic multi-service attack surface.

```
kshitz@kshitz:~$ sudo ss -tulnp
[sudo] password for kshitz:
Netid   State    Recv-Q   Send-Q       Local Address:Port      Peer Address:Port      Process
udp     UNCONN   0         0           127.0.0.54:53            0.0.0.0:*               users:((("systemd-resolve",pid=638,fd=16))
udp     UNCONN   0         0           127.0.0.53%lo:53         0.0.0.0:*               users:((("systemd-resolve",pid=638,fd=14))
udp     UNCONN   0         0           192.168.87.139%ens3:68   0.0.0.0:*               users:((("systemd-networkd",pid=618,fd=11))
tcp     LISTEN   0         4096        127.0.0.53%lo:53         0.0.0.0:*               users:((("systemd-resolve",pid=638,fd=15))
tcp     LISTEN   0         4096        0.0.0.0:22              0.0.0.0:*               users:((("sshd",pid=824,fd=3),("systemd",pid=1,fd=95))
tcp     LISTEN   0         151        127.0.0.1:3306          0.0.0.0:*               users:((("mysqld",pid=1099,fd=23))
tcp     LISTEN   0         4096        127.0.0.54:53          0.0.0.0:*               users:((("systemd-resolve",pid=638,fd=17))
tcp     LISTEN   0         70         127.0.0.1:33060         0.0.0.0:*               users:((("mysqld",pid=1099,fd=21))
tcp     LISTEN   0         4096        [::]:22                 [::]:*                  users:((("sshd",pid=824,fd=4),("systemd",pid=1,fd=96))
tcp     LISTEN   0         32         *:21                    *:*                      users:((("vsftpd",pid=829,fd=3))
tcp     LISTEN   0         511        *:80                    *:*                      users:((("apache2",pid=809,fd=4),("apache2",pid=808,fd=4),("apache2",pid=806,fd=4))
kshitz@kshitz:~$
```

✂ Attack Simulation

The goal of this phase was to generate realistic security events and verify that they are correctly detected by the Wazuh SIEM. Attack simulations were performed using Kali Linux (10.0.0.40) and suspicious activity was done manually using CMD and PowerShell in Windows 10 (John), and AD server respectively.

5.1 Domain-Level Attack Simulation (Active Directory)

The following PowerShell commands were executed on the Domain Controller:

```
$password = ConvertTo-SecureString "P@ssw0rd123!" -AsPlainText -Force
New-ADUser -Name "SOC_Test_User" `
-SamAccountName "soc_test" `
-UserPrincipalName "soc_test@kshitzcyberlab.local" `
-AccountPassword $password `
-Enabled $true
Add-ADGroupMember -Identity "Domain Admins" -Members "soc_test"
```

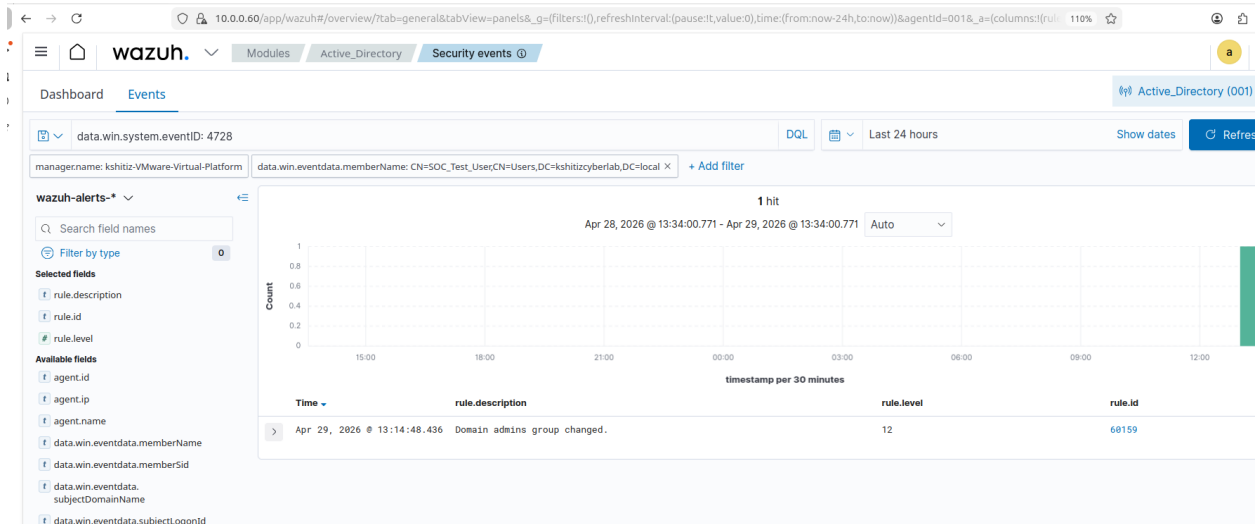
```
PS C:\Users\Administrator.WIN-MUTG0235J94> $password = ConvertTo-SecureString "P@ssw0rd123!" -AsPlainText -Force
>> New-ADUser -Name "SOC_Test_User" -SamAccountName "soc_test" -UserPrincipalName "soc_test@kshitzcyberlab.local" -AccountPassword $password -Enabled $true
PS C:\Users\Administrator.WIN-MUTG0235J94> Add-ADGroupMember -Identity "Domain Admins" -Members "soc_test"
PS C:\Users\Administrator.WIN-MUTG0235J94> gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
```

This activity simulates a real attack scenario where:

- A new user account is created inside the domain
- The user is immediately added to the **Domain Admins** group

This represents a **privilege escalation attack**, which is a common technique used after initial access.

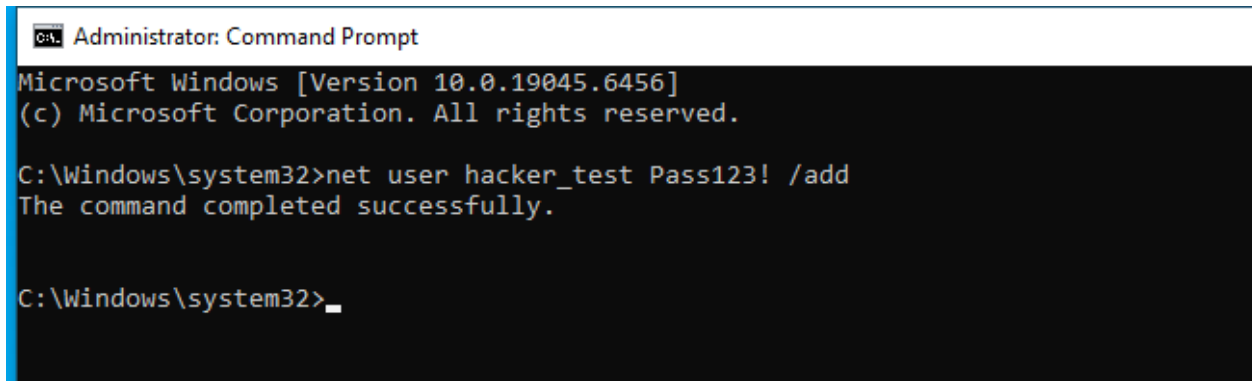


Wazuh - Wazuh	
10.0.0.60/app/wazuh#/overview/?tab=general&tabView=panels&_g=(filters:!((),refreshInterval:(pause:!t,value:0),time:(from:now-24h,to:now)))&_	
wazuh. Modules Active_Directory Security events	
data.win.eventdata.ogonid type	
data.win.eventdata.memberName	data.win.eventdata.targetUserName soc_test
data.win.eventdata.memberSid	data.win.system.channel Security
data.win.eventdata.newUacValue	data.win.system.computer DOMAIN_CONTROLLER_01.kshitizcyberlab.local
data.win.eventdata.oldUacValue	data.win.system.eventID 4738
data.win.eventdata.param1	data.win.system.eventRecordID 34729
data.win.eventdata.param2	data.win.system.keywords 0x8020000000000000
data.win.eventdata.param3	data.win.system.level 0
data.win.eventdata.param4	data.win.system.message > "A user account was changed."
data.win.eventdata.passwordLastSet	Subject: Security ID: S-1-5-7
data.win.eventdata.primaryGroupId	Account Name: ANONYMOUS LOGON
data.win.eventdata.processId	Account Domain: NT AUTHORITY
data.win.eventdata.processName	Account ID: 0x956
data.win.eventdata.queryName	data.win.system.opcode 0
data.win.eventdata.samAccountName	data.win.system.processID 1000
data.win.eventdata.serviceName	data.win.system.providerGuid {54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.eventdata.serviceSid	data.win.system.providerName Microsoft-Windows-Security-Auditing
data.win.eventdata.status	data.win.system.severityValue AUDIT_SUCCESS
data.win.eventdata.subjectDomainName	data.win.system.systemTime 2026-04-29T07:38:57.3312416Z
data.win.eventdata.subjectLogonId	
data.win.eventdata.subjectUserName	
data.win.eventdata.subjectUserSid	

5.2 Endpoint-Level Activity (Windows 10)

The following command was executed on the Windows 10 endpoint:

```
net user hacker_test Pass123! /add
```



```
C:\Windows\system32>net user hacker_test Pass123! /add
The command completed successfully.

C:\Windows\system32>
```

This simulates the creation of a local user account on an endpoint system.

This type of activity is often used by attackers to maintain persistence.

wazuh. Modules Windows-10-joh... Security events ⓘ						
Time ↓	Technique(s)	Tactic(s)	Description	Level	Rule ID	
> Apr 29, 2026 @ 13:04:23.141	T1087 T1059.003	Discovery, Execution	Suspicious Windows cmd shell execution	3	92032	
> Apr 29, 2026 @ 13:04:23.125	T1087 T1059.003	Discovery, Execution	Suspicious Windows cmd shell execution	3	92032	
> Apr 29, 2026 @ 13:04:23.079	T1059.003	Execution	Windows command prompt started by an abnormal process	4	92052	
> Apr 29, 2026 @ 13:04:20.641	T1087 T1059.003	Discovery, Execution	Suspicious Windows cmd shell execution	3	92032	
> Apr 29, 2026 @ 13:04:20.598	T1087 T1059.003	Discovery, Execution	Suspicious Windows cmd shell execution	3	92032	
> Apr 29, 2026 @ 13:04:20.597	T1059.003	Execution	Windows command prompt started by an abnormal process	4	92052	
> Apr 29, 2026 @ 13:04:14.294			Software protection service scheduled successfully.	3	60642	
> Apr 29, 2026 @ 13:04:12.484	T1484	Defense Evasion, Privilege Escalation	Users group changed.	5	60170	
> Apr 29, 2026 @ 13:04:12.453	T1098	Persistence	User account changed.	8	60110	
> Apr 29, 2026 @ 13:04:12.437	T1098	Persistence	User account enabled or created.	8	60109	

5.3 Attack Simulation (Kali Linux)

Attack simulations were performed from Kali Linux against the Ubuntu server (172.16.10.20).

Web Enumeration

```
gobuster dir -u http://172.16.10.20 -w /usr/share/wordlists/dirb/common.txt
```

This command attempts to discover hidden directories on the web server.

data.url	>	Apr 30, 2026 @ 17:49:28.131	Web server 400 error code.	5	31101
decoderftscmment	>	Apr 30, 2026 @ 17:49:28.130	Web server 501 error code (Not Implemented).	4	31121
decodername	>	Apr 30, 2026 @ 17:49:28.114	Web server 400 error code.	5	31101
decoderparent	>	Apr 30, 2026 @ 17:49:28.112	Web server 400 error code.	5	31101
full.log	>	Apr 30, 2026 @ 17:49:28.112	Web server 400 error code.	5	31101
id	>	Apr 30, 2026 @ 17:49:28.112	Web server 400 error code.	5	31101
inout.vtpe					

Network Scanning

`sudo nmap -sS -A 172.16.10.20`

This command performs service detection and port scanning.

```
└─$ sudo nmap -sS -A 172.16.10.20
[sudo] password for kshitiz:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-30 08:03 EDT
Stats: 0:00:47 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 98.12% done; ETC: 08:04 (0:00:00 remaining)
Nmap scan report for 172.16.10.20
Host is up (0.0022s latency).
Not shown: 988 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to ::ffff:10.0.0.40
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  At session startup, client count was 3
|_  vsFTPd 3.0.5 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.16 (Ubuntu Linux; protocol 2.0)
|_ssh-hostkey:
|_  256 be:ff:d9:60:f4:78:79:73:ef:f7:a3:d4:9c:52:13:72 (ECDSA)
|_  256 28:ae:98:0d:84:ca:a7:c9:17:7f:4c:9b:d3:03:58:04 (ED25519)
80/tcp    open  http      Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
255/tcp   filtered unknown
545/tcp   filtered ekshell
1183/tcp  filtered llsurfup-http
2222/tcp  filtered EtherNetIP-1
2604/tcp  filtered ospfd
3031/tcp  filtered eppc
7777/tcp  filtered cbt
18040/tcp filtered unknown
32770/tcp filtered sometimes-rpc3
Device type: general purpose
Running: Linux 4.X
OS CPE: cpe:/o:linux:linux_kernel:4
OS details: Linux 4.19 - 5.15
Network Distance: 2 hops
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE (using port 143/tcp)
HOP RTT ADDRESS
1 1.31 ms 10.0.0.1
2 1.55 ms 172.16.10.20

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 47.91 seconds
```


The screenshot shows the Wazuh Security events interface. The top navigation bar includes the Wazuh logo, a dropdown menu, and tabs for Modules, SSH-Ubuntu, and Security events. The main table displays a list of security events for agent 10.0.0.60. The table has four columns: Time, rule.description, rule.level, and rule.id. The events are sorted by time, showing a sequence of FTP session openings and authentication successes, followed by several 400 error codes from the web server.

Time	rule.description	rule.level	rule.id
> Apr 30, 2026 @ 17:49:38.166	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:38.166	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:38.124	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP Authentication success.	3	11402
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP Authentication success.	3	11402
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP session opened.	3	11401
> Apr 30, 2026 @ 17:49:28.173	vsftpd: FTP Authentication success.	3	11402
> Apr 30, 2026 @ 17:49:28.131	Web server 400 error code.	5	31101
> Apr 30, 2026 @ 17:49:28.130	Web server 501 error code (Not Implemented).	4	31121
> Apr 30, 2026 @ 17:49:28.114	Web server 400 error code.	5	31101
> Apr 30, 2026 @ 17:49:28.112	Web server 400 error code.	5	31101
> Apr 30, 2026 @ 17:49:28.112	Web server 400 error code.	5	31101

SSH Brute Force Behavior

Multiple login attempts were generated against the SSH service, resulting in authentication failures.

```
File Actions Edit View Help
$ hydra -l kshitiz -P /usr/share/wordlists/metasploit/unix_passwords.txt ssh://172.16.10.20 -t 4 -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-30 07:59:46
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1021 login tries (l:1/p:1021), ~256 tries per task
[DATA] attacking ssh://172.16.10.20:22/
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "admin" - 1 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "123456" - 2 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "12345" - 3 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "123456789" - 4 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "password" - 5 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "iloveyou" - 6 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "princess" - 7 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "1234567" - 8 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "12345678" - 9 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "abc123" - 10 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "nicole" - 11 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "daniel" - 12 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "babygirl" - 13 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "monkey" - 14 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "lovely" - 15 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "jessica" - 16 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "654321" - 17 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "michael" - 18 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "ashley" - 19 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "qwerty" - 20 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "111111" - 21 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "iloveu" - 22 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "000000" - 23 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "michelle" - 24 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "tigger" - 25 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "sunshine" - 26 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "chocolate" - 27 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "password1" - 28 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "soccer" - 29 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "anthony" - 30 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "friends" - 31 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "butterfly" - 32 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "purple" - 33 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "angel" - 34 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "jordan" - 35 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "liverpool" - 36 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "justin" - 37 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "loveme" - 38 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "fuckyou" - 39 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "123123" - 40 of 1021 [child 0] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "football" - 41 of 1021 [child 2] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "secret" - 42 of 1021 [child 3] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "andrea" - 43 of 1021 [child 1] (0/0)
[ATTEMPT] target 172.16.10.20 - login "kshitiz" - pass "carlos" - 44 of 1021 [child 0] (0/0)
```

wazuh. Modules SSH-Ubuntu Security events				
>	Apr 30, 2026 @ 17:48:04.001	Maximum authentication attempts exceeded.	8	5758
>	Apr 30, 2026 @ 17:48:03.999	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:48:03.997	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:48:03.996	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:48:02.000	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:48:01.997	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:48:01.995	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:48:01.994	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:47:59.999	PAM: User login failed.	5	5503
>	Apr 30, 2026 @ 17:47:59.994	syslog: User missed the password more than one time	10	2502
>	Apr 30, 2026 @ 17:47:59.992	syslog: User authentication failure.	5	2501
>	Apr 30, 2026 @ 17:47:59.992	Maximum authentication attempts exceeded.	8	5758
>	Apr 30, 2026 @ 17:47:57.995	sshd: brute force trying to get access to the system. Authentication failed.	10	5763
>	Apr 30, 2026 @ 17:47:57.992	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:47:57.990	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:47:57.989	sshd: authentication failed.	5	5760
>	Apr 30, 2026 @ 17:47:56.030	sshd: authentication failed.	5	5760

5.4 Detection Results in Wazuh

The Wazuh SIEM successfully detected all simulated activities.

Detection of local user account creation on Windows endpoint

SSH Attack Detection

- Multiple authentication failed logs
- Maximum authentication attempts exceeded
- PAM: User login failed
- Indicators of brute-force attempts

Web Attack Detection

- HTTP 400 and 501 error responses
- Suspicious request patterns from enumeration

Port Scanning Detection

- FTP session creation logs

Summary

The SIEM successfully collected and correlated these events, proving that the monitoring setup is functioning correctly.

⚠ Lab Limitation

Due to hardware limitations (16 GB RAM), all systems could not run simultaneously.

❏ Firewall Rules

NAME	TAGS	TYPE	ZONE	Source	Destination	APPLICATION	SERVICE	ACTION	PROFILE	OPTIONS	HIT COUNT
1 Allow Server-Internet	none	universal	any	any	any	any	any	Allow	none		8357
2 Allow Windows 10 to AD server	none	universal	any	10.0.0.0/24	any	any	any	Allow	none		161
3 Allow AD server to Wazuh	none	universal	any	172.16.10.10	any	any	any	Allow	none		0
4 Allow VPN-Access	none	universal	any	10.0.0.0/24	any	any	any	Allow	none		5
5 Allow VPN-RDP-AD	none	universal	any	10.0.0.0/24	any	any	any	Allow	none		0
6 Kali SSH Ubuntu	none	universal	any	10.0.0.0/24	any	any	any	Allow	none		1120
7 SSH Ubuntu to Wazuh	none	universal	any	172.16.10.10	any	any	any	Allow	none		0
8 Intrazone-default	none	intrazone	any	any	any	any	any	Allow	none		15014
9 Interzone-default	none	interzone	any	any	any	any	any	Deny	none		28

Results

The lab successfully demonstrates:

- Centralized log collection using SIEM
- Detection of identity-based attacks
- Detection of brute-force and scanning activity
- Network segmentation using a firewall
- Visibility across multiple log sources

Hardware Allocation

System	RAM
pfSense	1 GB
Palo Alto	5.5 GB
Wazuh (Ubuntu)	4 GB
Windows 10	2.1 GB
Windows Server 2022	2.6 GB
Kali Linux	2 GB
Ubuntu Server	1 GB

Key Takeaways

This project demonstrates practical understanding of:

- SIEM deployment and log analysis
- Active Directory monitoring
- Firewall-based segmentation
- Attack simulation techniques
- Security event detection
- Writing firewall rules

References

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